



core EMG

Quick Start Guide

On behalf of Noraxon...

Welcome and congratulations on acquiring your new Core EMG System!

This guide will provide you with step-by-step instructions on how to install your new hardware and software, adjust device settings, and record your first measurement.

Let's begin by walking through how to install your new hardware.

Note: This is not meant to be a complete user manual, but a guide to help you get started with your system. Refer to the Ultium EMG User Manual (#P-8848) for more detailed operating instructions for the Core EMG System.

1 System Unboxing

The Core EMG System is packed within a reinforced padded box for storage and protection during transport. Upon receiving the package, carefully remove all contents and verify the following components are present.



Figure 1 – Core Hub (#884)



Figure 2 – Core EMG Sensor (#816)



Figure 3 – 5V USB-C Power Supply & Cable (#PSU11 & #CBL44)

Additional items that may be included with your Core EMG System include:

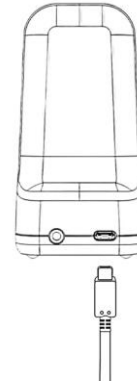
- Dual Solid Gel EMG Electrodes (#271S)
- Core EMG User Manual (#P-8848)

2 Installing the Hardware

Step 1

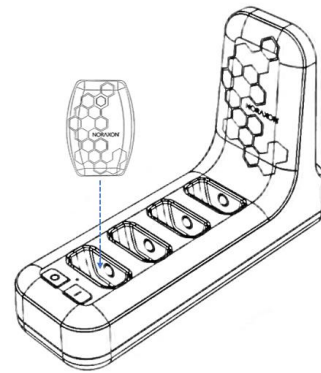
Insert the USB-C cable (#CBL44) into the USB connector on the back of the Core Hub (#884).

Insert the opposite end of the USB-C cable into an available USB-C port on the computer.



Step 2

Insert the Core EMG Sensors (#816) into the Hub.



3 Installing the Companion Software - myoRESEARCH™

To utilize the full functionality of the Core EMG system, and ensure the system has updated drivers, Noraxon's *myoRESEARCH* (MR) needs to be installed on the computer.

Note: The Core Hub requires the Noraxon USB device driver which is pre-installed by the MR3 software installation. It is also available in the [Downloads](#) section of the Noraxon website.

3.1 Software Installation

Within the Core EMG System shipment, there is a USB flash drive containing the latest *myoRESEARCH* software.

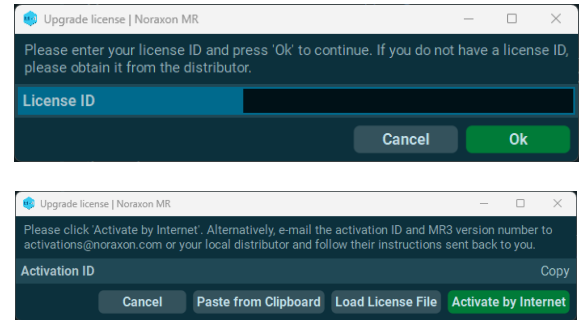
1. Insert the MR USB flash drive into the PC
2. Open the installer application (e.g., noraxon.mr.4.0.0) and follow the instructions
3. After installation, an icon will be created on the desktop



3.2 Companion Software Activation

The companion software must be *activated* before unrestricted use is possible.

1. Open myoRESEARCH
2. A dialog box will indicate how many more times MR can be opened in trial mode
3. Click **Activate**
4. Enter the License ID provided on your USB flash drive and click **Ok**
5. If you have an internet connection, click **Activate by Internet** for immediate activation
 - a. Alternatively, email the provided activation ID to activation@noraxon.com
 - b. Noraxon Support will email or respond by phone with the Activation Code
 - c. Enter the provided Activation Code to remove any restrictions on use

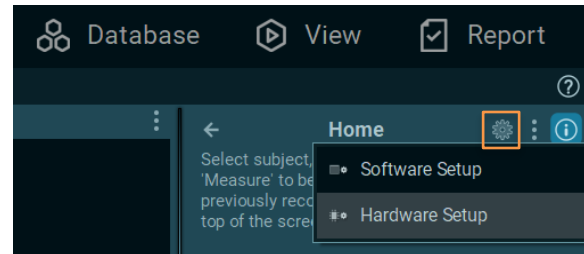


4 Configuring the Hardware

Before the Core EMG system can be used, the device software settings must be configured to recognize the different components that make up the system. Follow the instructions below to update the hub firmware, sensor firmware, and populate sensors to prepare for data collection.

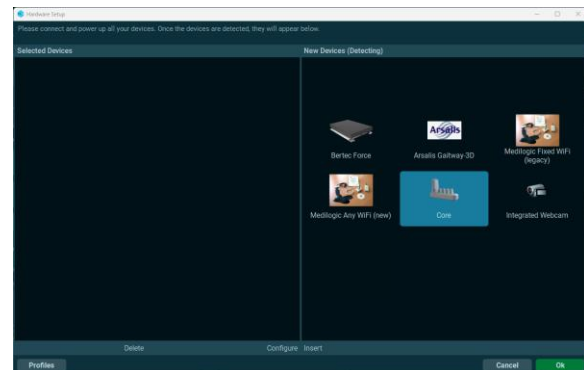
Step 1

Open the MR software, click on the settings icon (⚙️) in the upper right-hand corner, then click **Hardware Setup**.



Step 2

After connecting the Core Hub to a USB port on the computer, select the Core Hub icon in the **New Devices** area, and click **Insert**.



Step 3

The Core EMG hardware setup window will appear as shown.

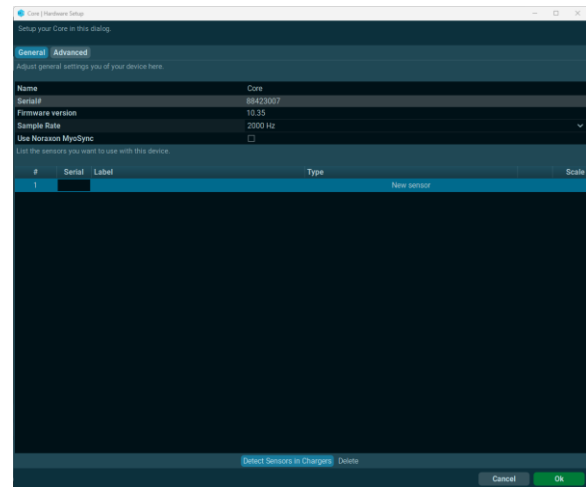
Within the “General” tab, select the desired settings for the system, i.e., name, sample rate.

Note: If available, firmware updates for the Receiver and Sensors will be indicated here.

If using myoSYNC, click the **Use Noraxon myoSYNC** checkbox.

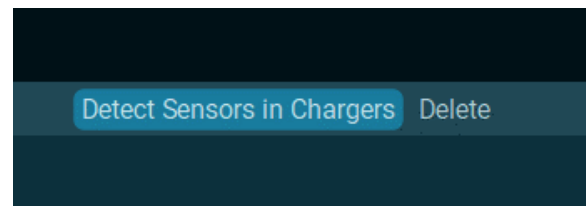
Additional settings, i.e., RF network, are available within the “Advanced” tab.

Note: If multiple Core or Ultium systems are in use, separate RF network assignments are required for each Receiver.

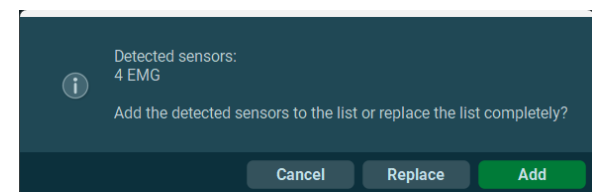
**Step 4**

Make sure the Core EMG Sensors are placed in the Hub, then click **Detect Sensors in Chargers** to load all sensor serial numbers into the MR software.

Note: The first serial number in the list is associated with the Sensor closest to the front of the Charger (i.e., toward the Charger LEDs).

**Step 5**

After detection is complete, click **Replace** to replace all existing sensor serial numbers associated with this device or **Add** to append additional sensor serial numbers to an already existing list.



List the sensors you want to use with this device.				
#	Serial	Label	Type	
1	40013	EMG 1	EMG	
2	40014	EMG 2	EMG	
3	40019	EMG 3	EMG	
4	40029	EMG 4	EMG	

Step 6

Click **Ok** to save the hardware settings and close the setup window.



5 Recording a Measurement

Step 1

On the **Home** screen, select or create a subject.

Select an application, then select or create a new protocol.

Click **MEASURE** to proceed.

Step 2

Insert the Core EMG device into the configuration by dragging it from the **Available Devices** list to the **Devices in Your Configuration** area.

Step 2

Select the channels to include in the measurement by clicking the checkmark.

While a channel is selected, assign a label by selecting the corresponding muscle on the **Muscle Map** or entering the name manually.

Click **MEASURE** to proceed.

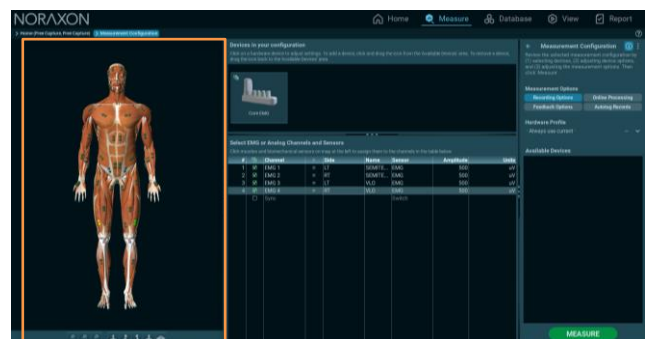
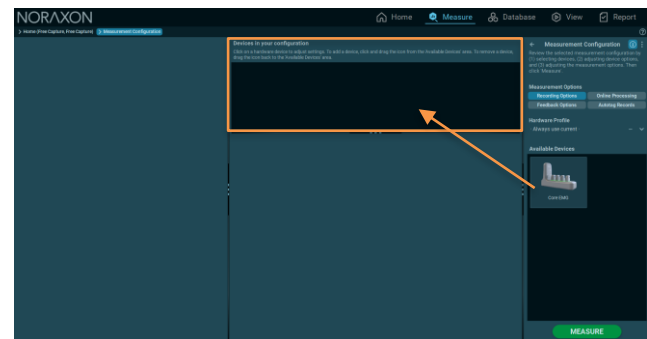
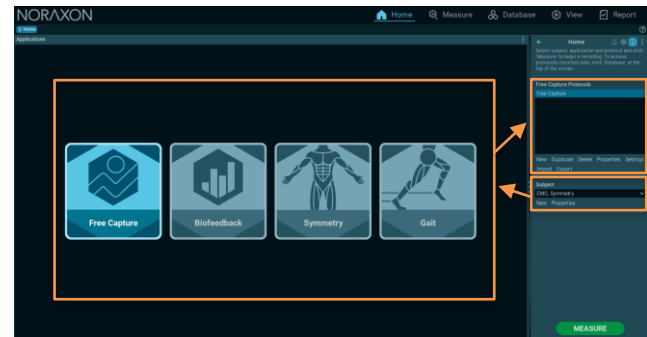
Step 4

Confirm signal quality and click **RECORD** to begin your recording data.

Step 5

Click **STOP** to end the recording, then enter a name for the new record.

Click **Save & View** to view and begin processing the record, or one of the other options (**Discard & Measure Again** or **Save & Measure Again**) to return to the measurement screen.



Save Data

Please enter name for the new record.

Name

Free Capture

Subject

1, Subject

New

Discard & Measure Again

Save & Measure Again

Save & View

6 Viewing a Record

To view a previously recorded measurement, click the **Database** tab. Records are organized by subject name and date measured. Select and click **VIEW** or double-click on the record of interest to open the record in the **View** tab.

7 Further Use Features of MR

There are many additional features built within MR, including:

- Multi-activity recordings
- Live biofeedback visual displays
- Online and post-measurement signal processing
- Customized reporting
- Exporting (and importing) of data

To learn more about the features available to you through the system(s) you have purchased, refer to the corresponding Hardware User Manual for this device (#P-8848).

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